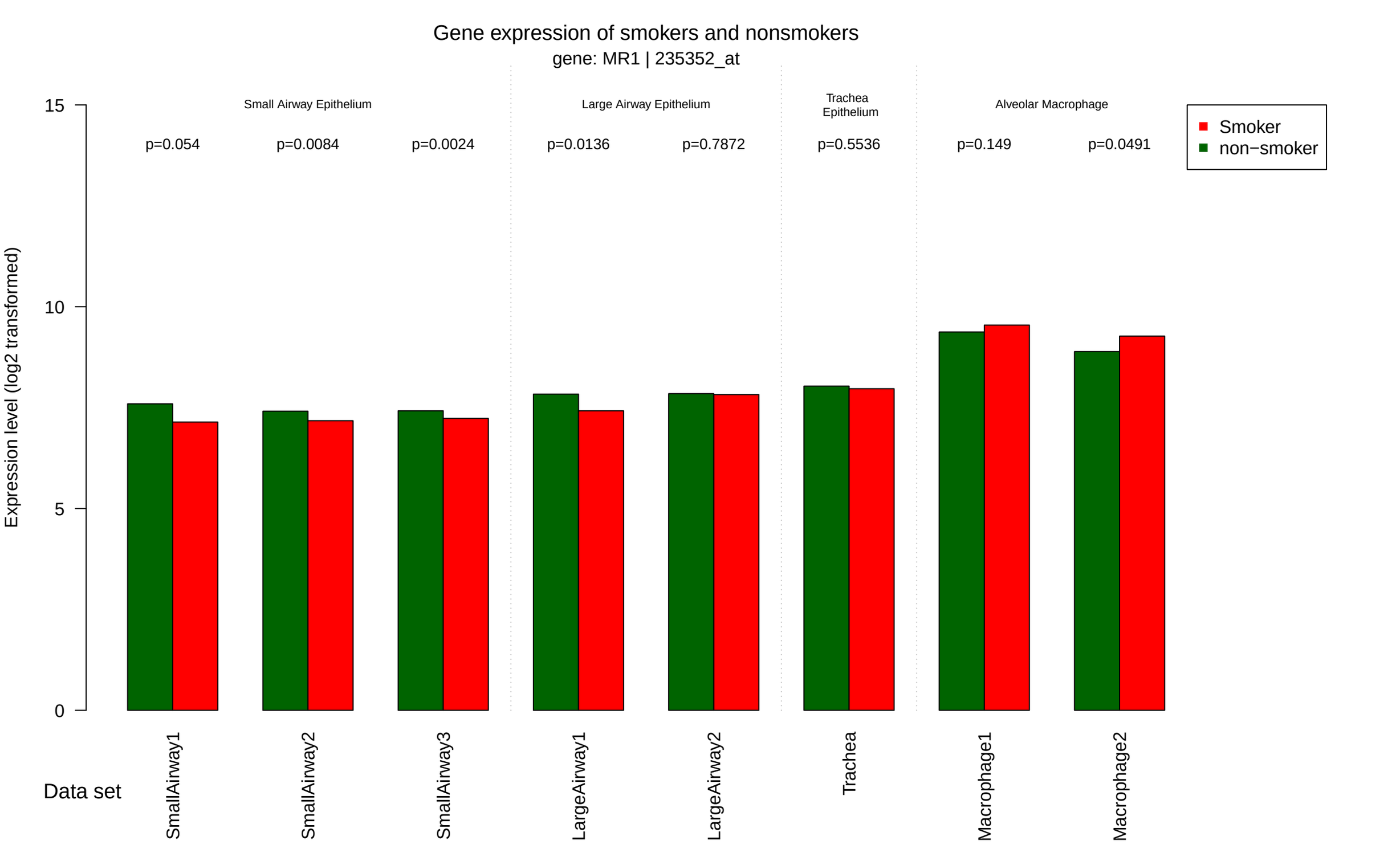
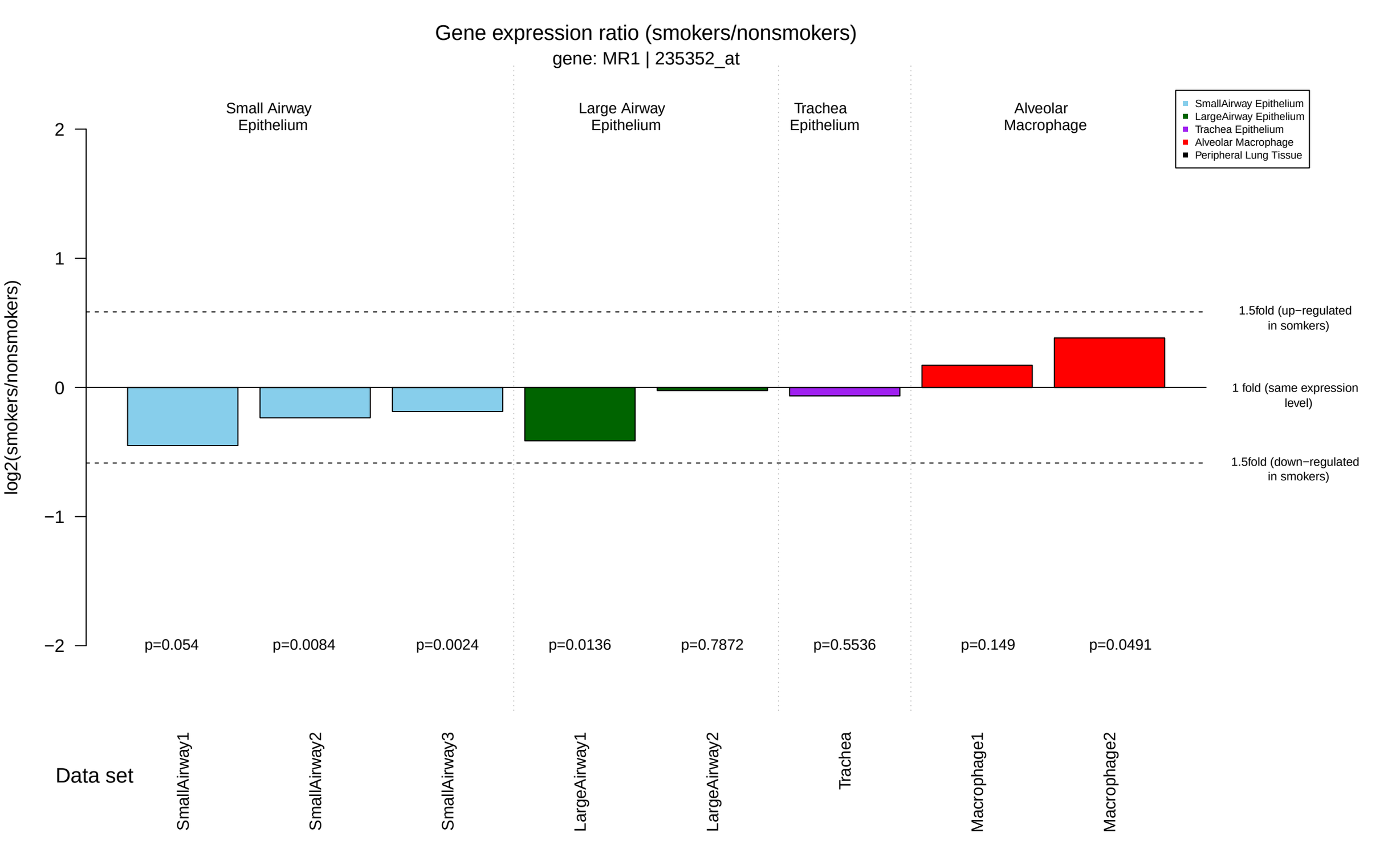
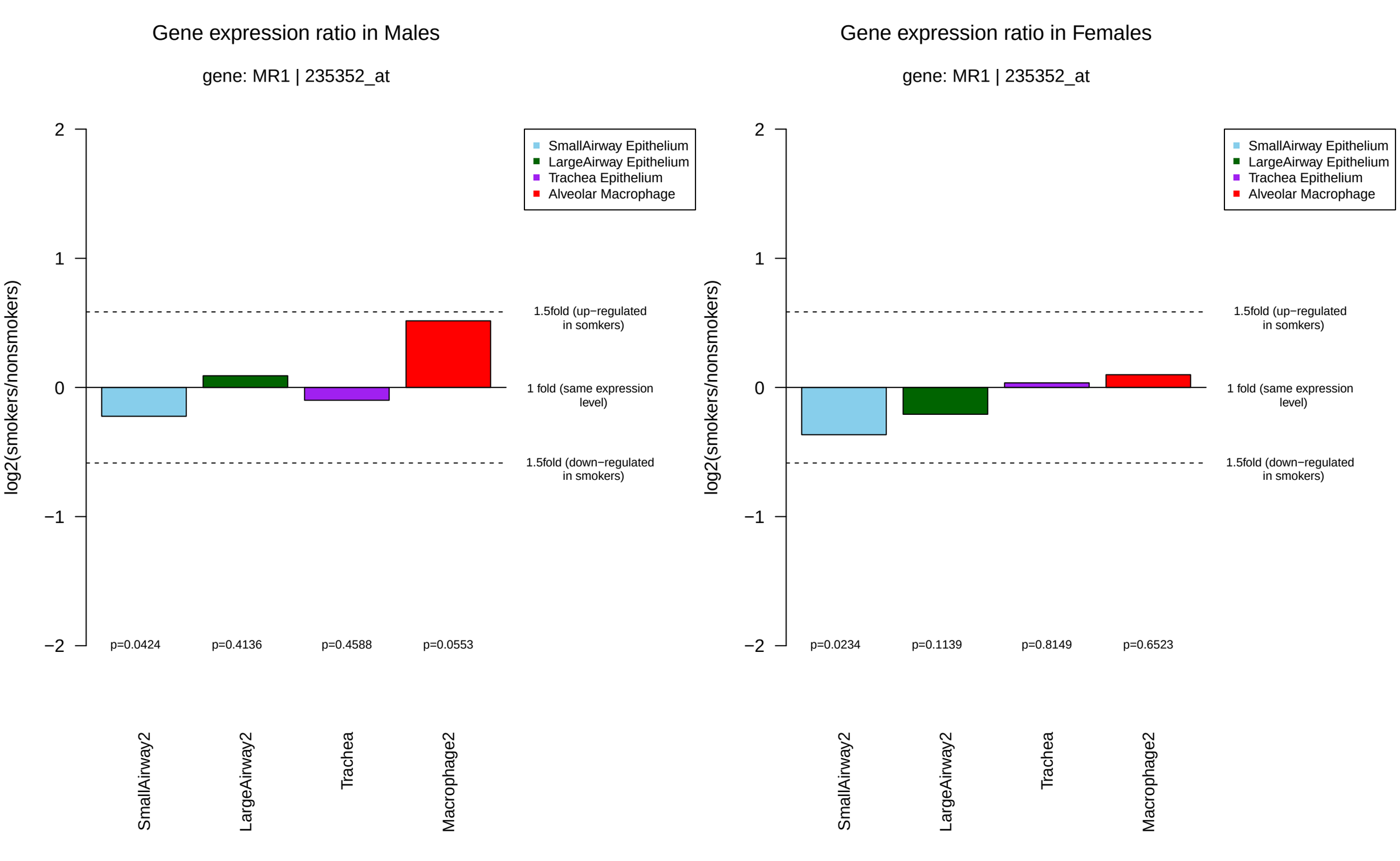
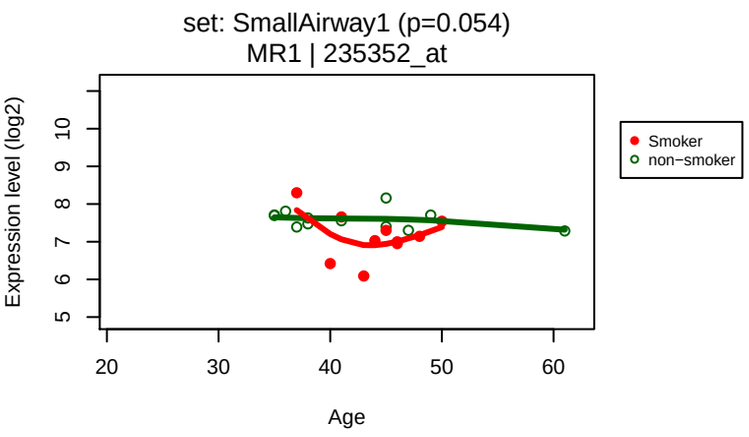
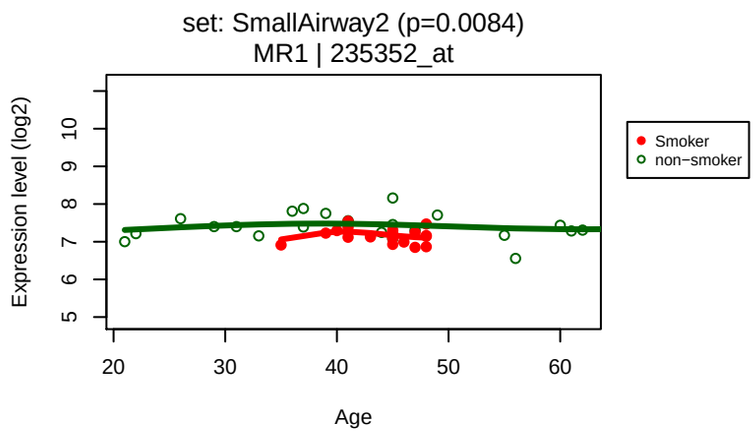
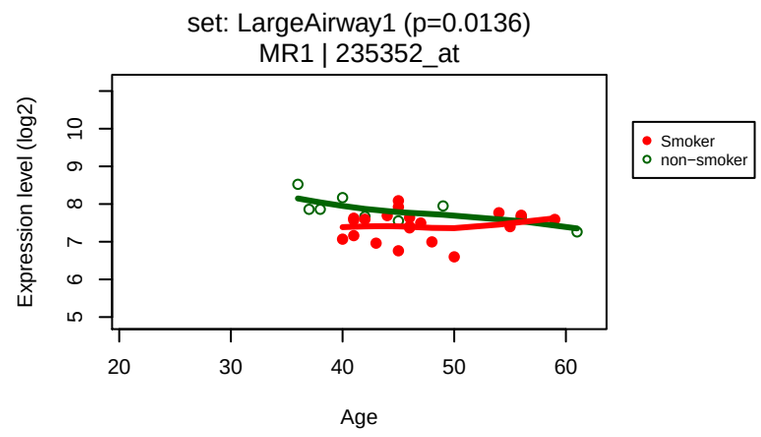

Gene expression of smokers and nonsmokers
gene: MR1 | 235352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.054
p=0.0084
p=0.0024
p=0.0136
p=0.7872
p=0.5536
p=0.149
p=0.0491
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

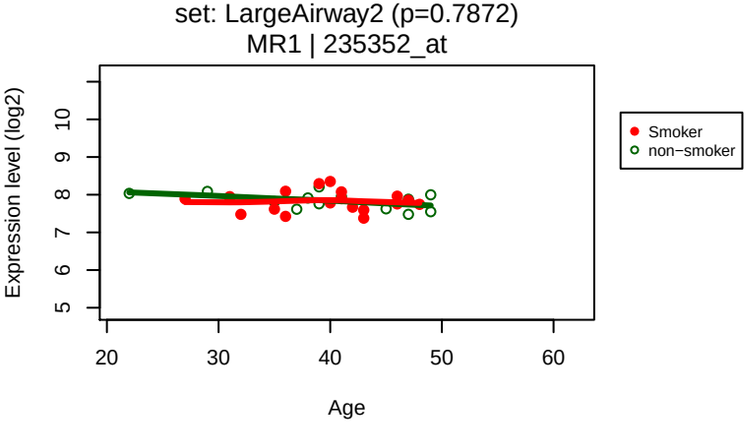

Gene expression ratio (smokers/nonsmokers)
gene: MR1 | 235352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.054
p=0.0084
p=0.0024
p=0.0136
p=0.7872
p=0.5536
p=0.149
p=0.0491
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

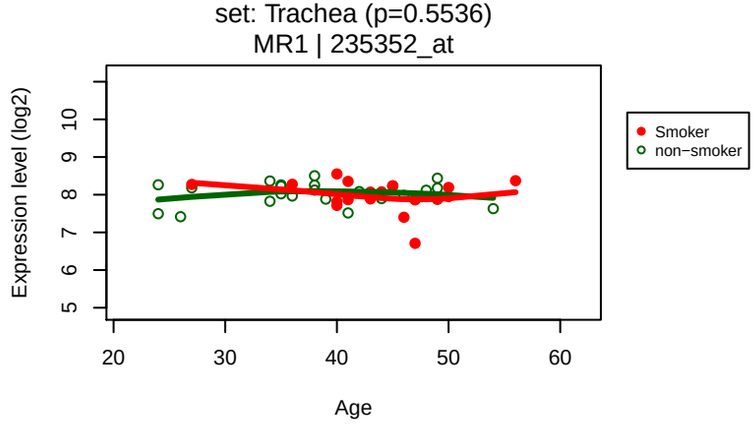

Gene expression ratio in Males
gene: MR1 | 235352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0424
p=0.4136
p=0.4588
p=0.0553
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MR1 | 235352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0234
p=0.1139
p=0.8149
p=0.6523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

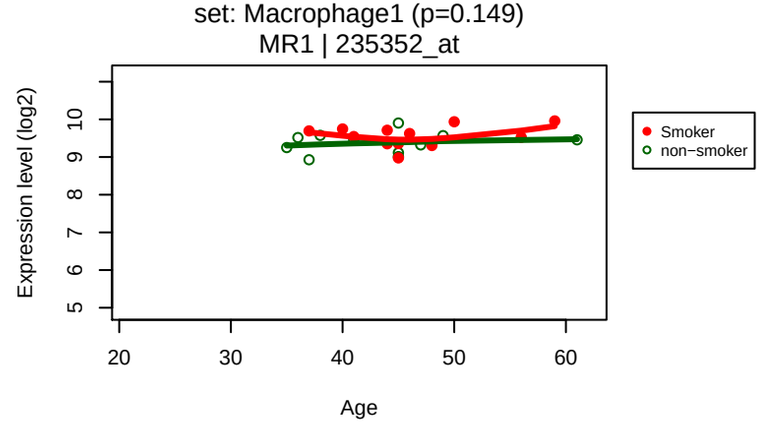

set: SmallAirway1 (p=0.054)
MR1 | 235352_at
Expression level (log2)
Age
Smoker
non−smoker

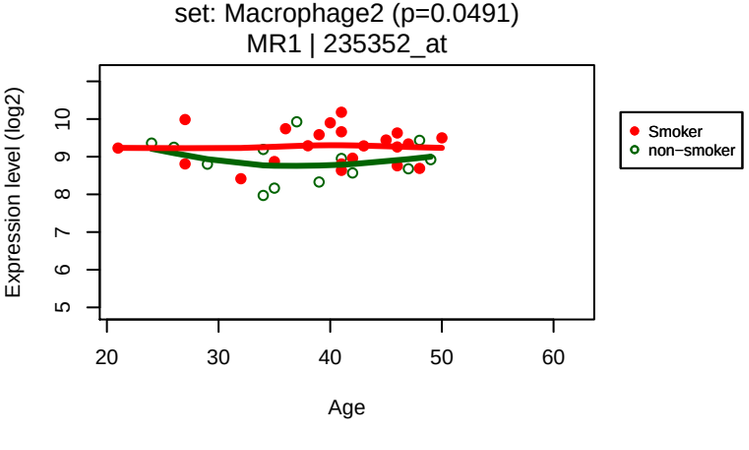

set: SmallAirway2 (p=0.0084)
MR1 | 235352_at
Expression level (log2)
Age
Smoker
non−smoker

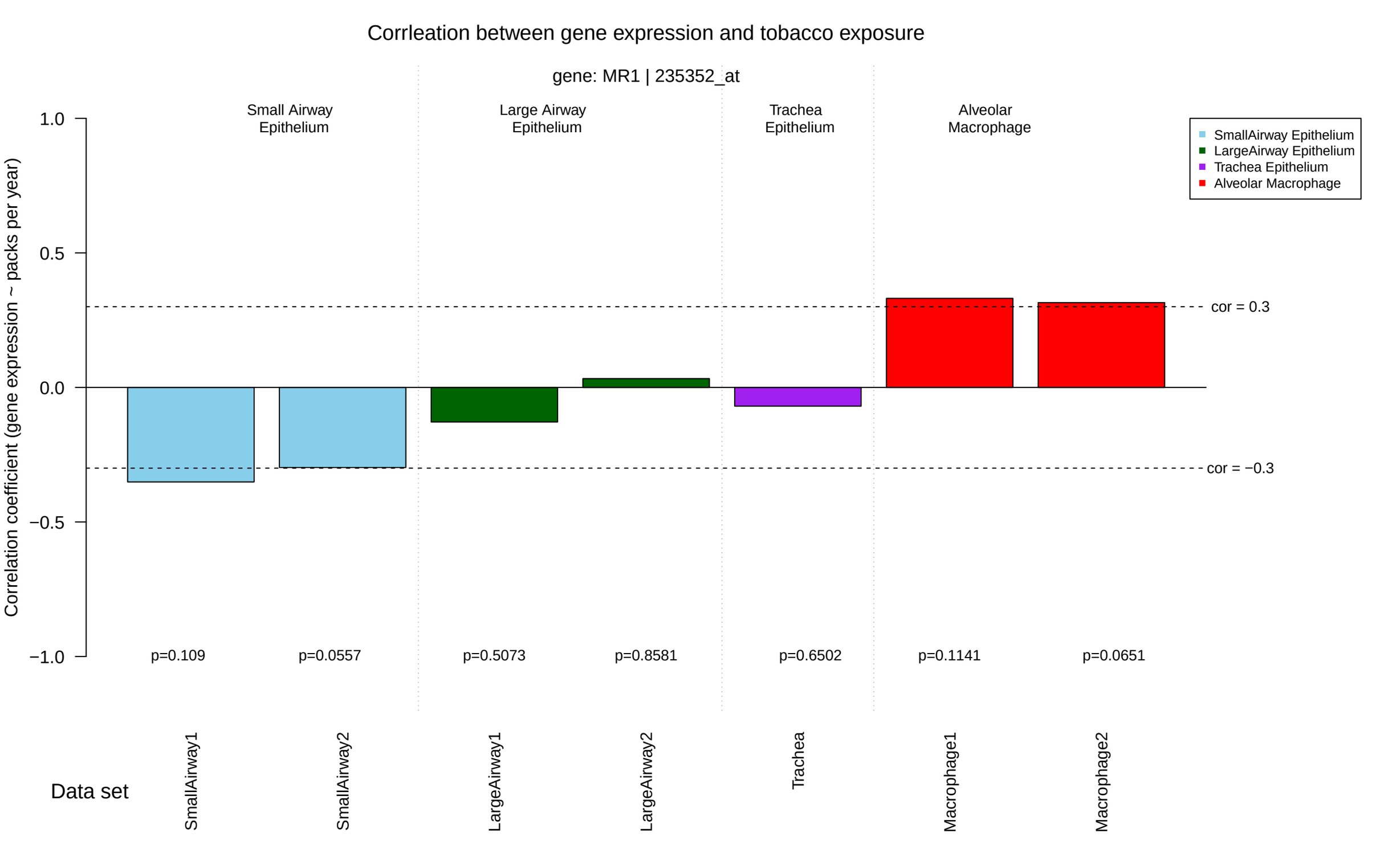

set: LargeAirway1 (p=0.0136)
MR1 | 235352_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7872)
MR1 | 235352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5536)
MR1 | 235352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.149)
MR1 | 235352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0491)
MR1 | 235352_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MR1 | 235352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.109
p=0.0557
p=0.5073
p=0.8581
p=0.6502
p=0.1141
p=0.0651
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2